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2)

Intercepted Suspicion Message

a) Original message:

The coded message is

"GBRH RHZ HWXVVG"

I + used the Shift cipher, when each letter is replaced by its seven alphabet equivalent

• $G \rightarrow T$

• $S \rightarrow H$

• $R \rightarrow I$

• $H \rightarrow S$

$n \rightarrow I$

$Z \rightarrow K$

$H \rightarrow S$

$V \rightarrow E$

$x \rightarrow C$

$I \rightarrow n$

$V \rightarrow E$

$G \rightarrow T$

Therefore the original message

"This is A Secret"

Interpretation:

The message has word pattern of
4, -2, -1, 6 which match the structure of
"This is A secret. He expected 'RH', represents
"is" showing a constant substitution

The start cipher is a simple substitution
cipher in which $A \rightarrow Z$, $B \rightarrow Y$, $C \rightarrow X$, and
so on, hence "CRSRHRH Z HWVVG"
decrypts to "This is secret."

Secure message system:

a) Encryption of "HELLO" using sequence
"Key"
using the maximum value repeat
the keyword to match the message length.

Message: HELLO

Keyword: KEYKE

using $A=0, B=1, Z=25$.

Character	H	E	L	L	O
message value	7	4	11	11	14
Key	K	E	Y	K	E
Key value	10	4	24	10	4
sum mod 26	17	8	9	21	18
C alphabet	R	I	J	V	S

= RIJVS

Therefore :

HELLO $\rightarrow$ RIJVS.

b) Effect the repeating keyword.

The key words.. "KEY" is repeated as
"FEYKE" so the each character of the
message has a corresponding key character.
each plaintext letter is shifted according
to corresponding keyword letter.

Therefore - the same plaintext letter
can produce different ciphertext letters
depending on the key character used.